

# Sampling contacts from sequence

A figure-first writeup on existing predictors, oracle contacts, MarinFold and useful sampling diversity.

Fixed model: exp277 step 266344, 1.47B parameters, 248.584B raw training tokens. Natural proteins lead: 97 validation + 217 publication test. Designs remain separate. AF2/AF3/Boltz-2 comparators use shared archived MSAs and no templates, with confidence selection.

Predictor generation → cached analysis tables → fast static and interactive rendering. Every numerical figure element maps back to a source CSV row.

# What the contact and confidence analyses show

AF2 / AF3 / Boltz-2 GDT-TS: 0.854 / 0.860 / 0.861 on 305 natural proteins. At MSA depth <10: 0.196 / 0.332 / 0.239, with only five proteins.

MarinFold contact R-precision: 0.561 overall; 0.345 at MSA depth <10 versus 0.617 at  $\geq 1000$ . Only five natural proteins occupy the shallowest tier.

Oracle confidence ranks 1, 5, 1, 5, 1 out of 101 maps for the five low-depth proteins. 100 ESMFold2 maps/protein, a shared eligible-pair mask, three Helico samples/map. Original ESMFold2 median GDT-TS: 0.537, 0.718, 0.699, 0.315, 0.725. Confidence-source GDT-TS Spearman:  $-0.041$ ,  $-0.064$ ,  $0.078$ ,  $-0.049$ ,  $0.556$ . Protein order: 8ii8\_A, 8oxk\_A, 8qoh\_A, 8ux2\_A, 8wrx\_A. Only five biological examples.

Top-L contacts raise Helico GDT-TS from 0.150 to 0.504 on 305 matched natural proteins. The paired gain is 0.354 [0.323, 0.387]. All cuts were fixed before test inference.

# Useful diversity is the open question

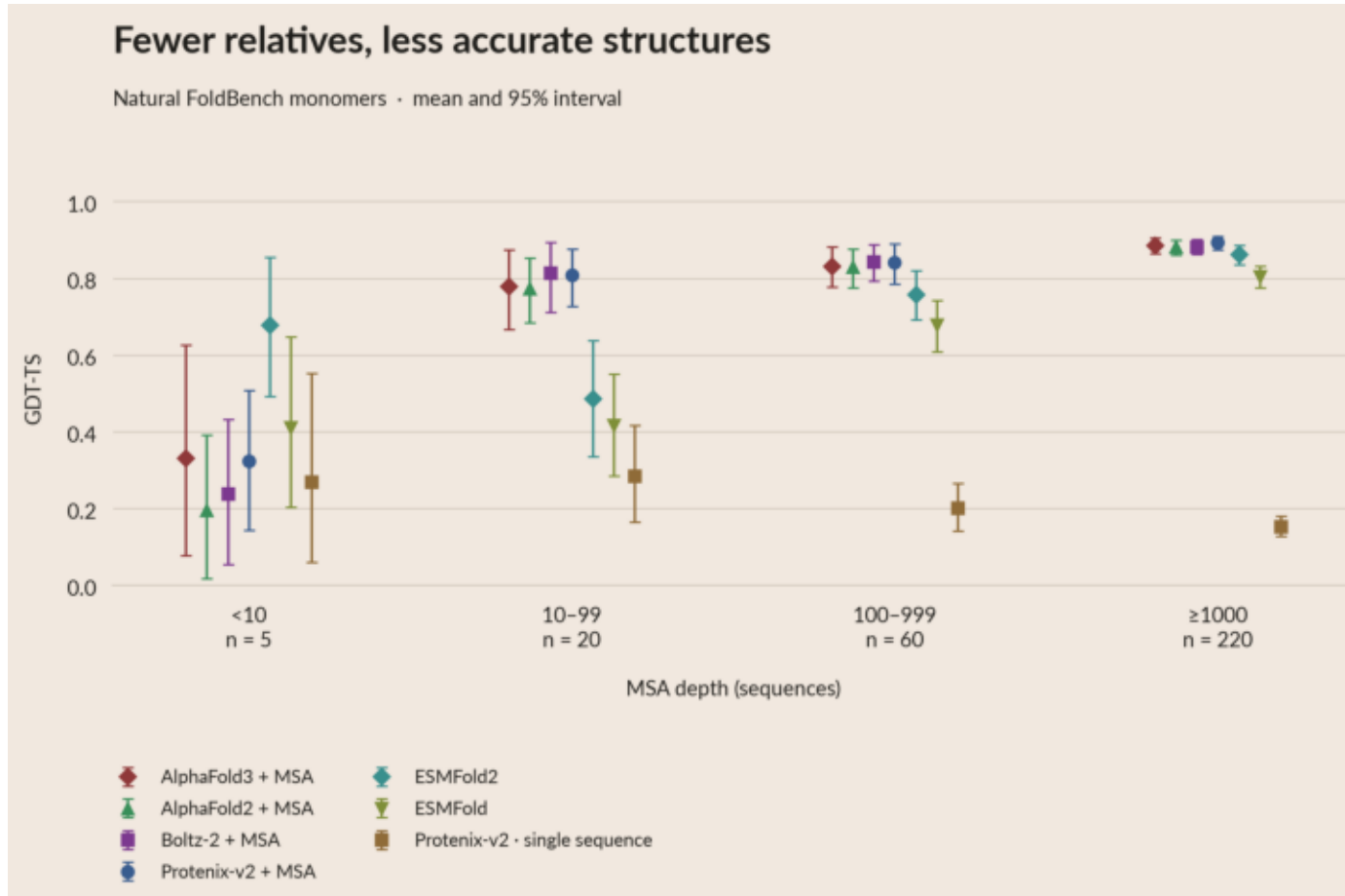
314 proteins  $\times$  100 samples: consensus R-precision 0.561; oracle best individual map 0.528. Paired difference  $-0.032$ , 95% bootstrap interval  $[-0.038, -0.025]$ .

All 100 maps are distinct for every protein; mean Jaccard 0.277, true-contact union recall 0.943. This is not an absence of contact-set diversity, and it does not directly measure distinct folds.

Next: better whole-map candidates, inference-time search and post-training.

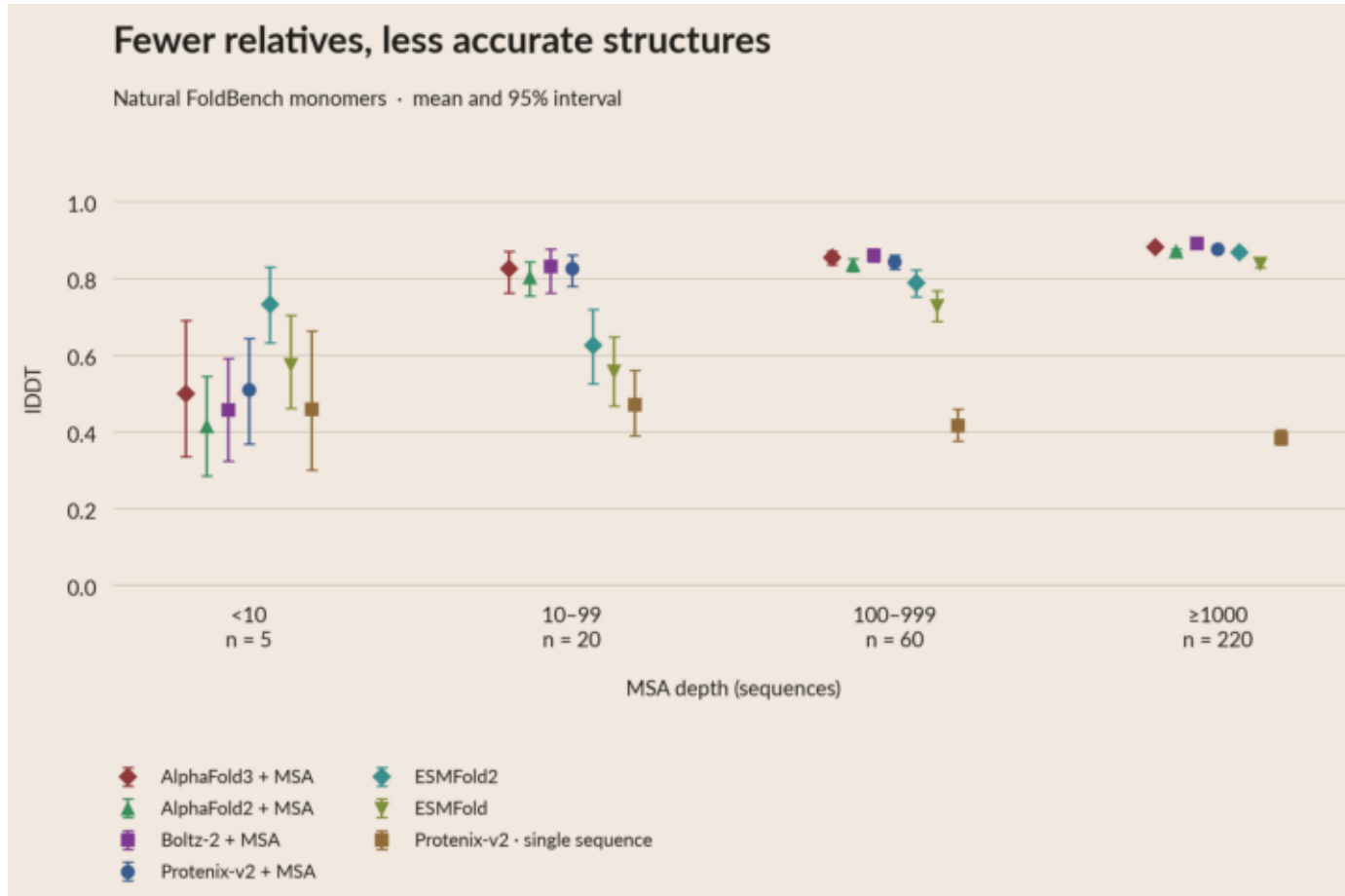
# 01\_predictors.png

Natural FoldBench monomers; the same 305 proteins in every structural arm. Points are protein means; bars are 95% protein-bootstrap intervals. MSA depth counts sequences, including the query, in the alignment used by Protenix-v2 + MSA. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



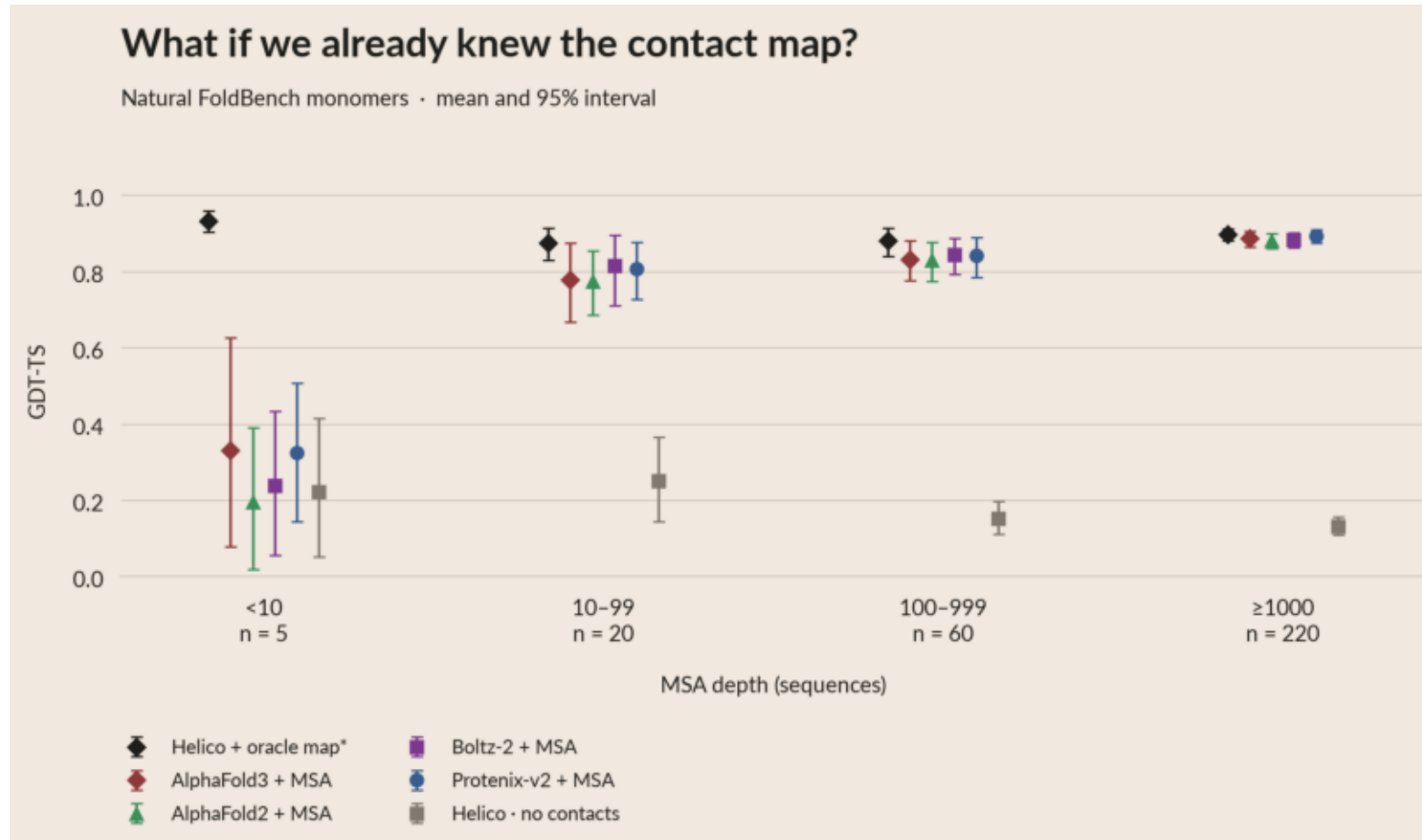
# 01\_predictors\_lddt.png

Natural FoldBench monomers; the same 305 proteins in every structural arm. Points are protein means; bars are 95% protein-bootstrap intervals. MSA depth counts sequences, including the query, in the alignment used by Protenix-v2 + MSA. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



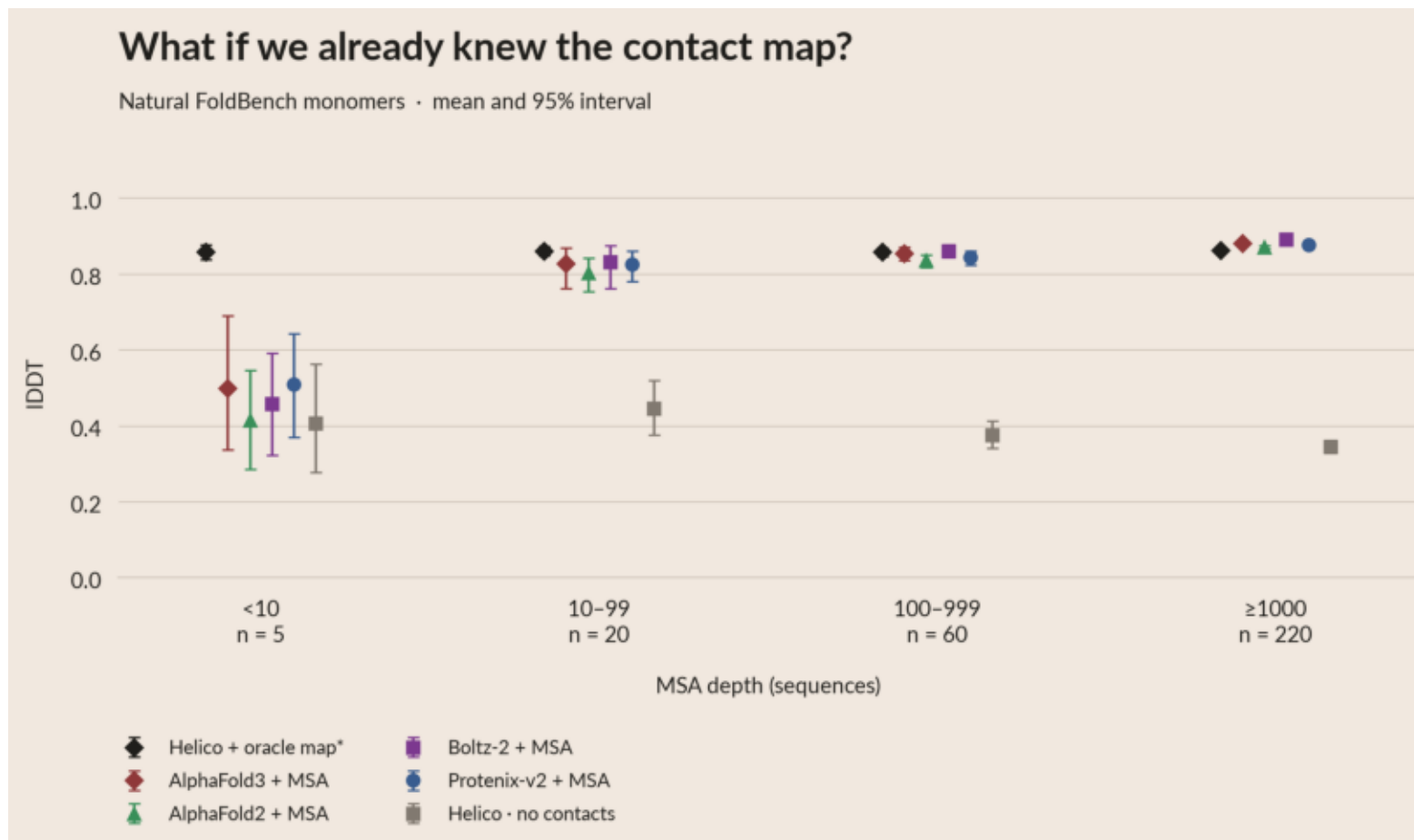
## 02\_oracle.png

\*Oracle = ground-truth contacts AND non-contacts supplied to Helico without an MSA. This is an information upper bound, not a competing predictor or a matched-count true-contact experiment. Same 305 natural proteins as Figure 1. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



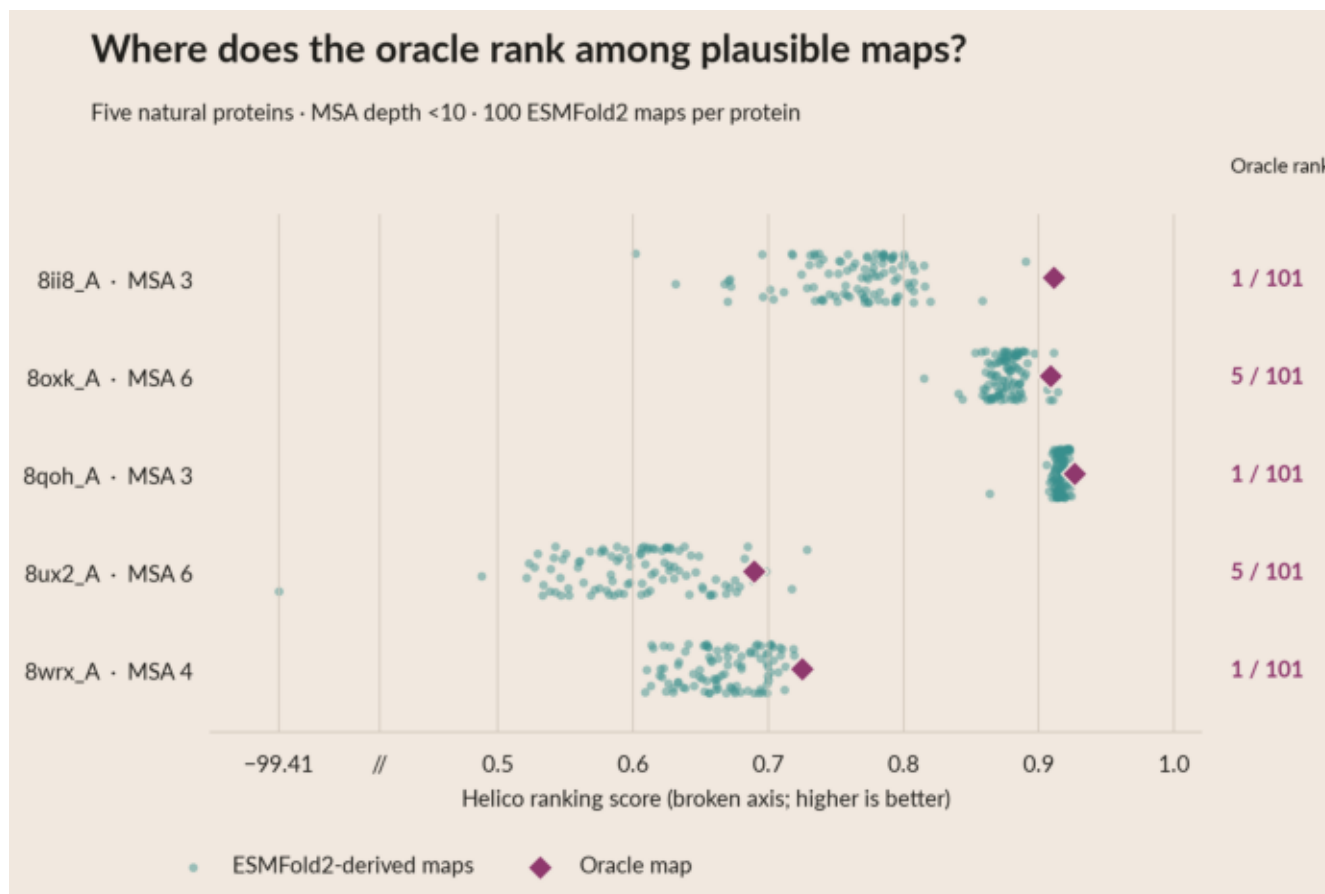
## 02\_oracle\_lddt.png

\*Oracle = ground-truth contacts AND non-contacts supplied to Helico without an MSA. This is an information upper bound, not a competing predictor or a matched-count true-contact experiment. Same 305 natural proteins as Figure 1. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



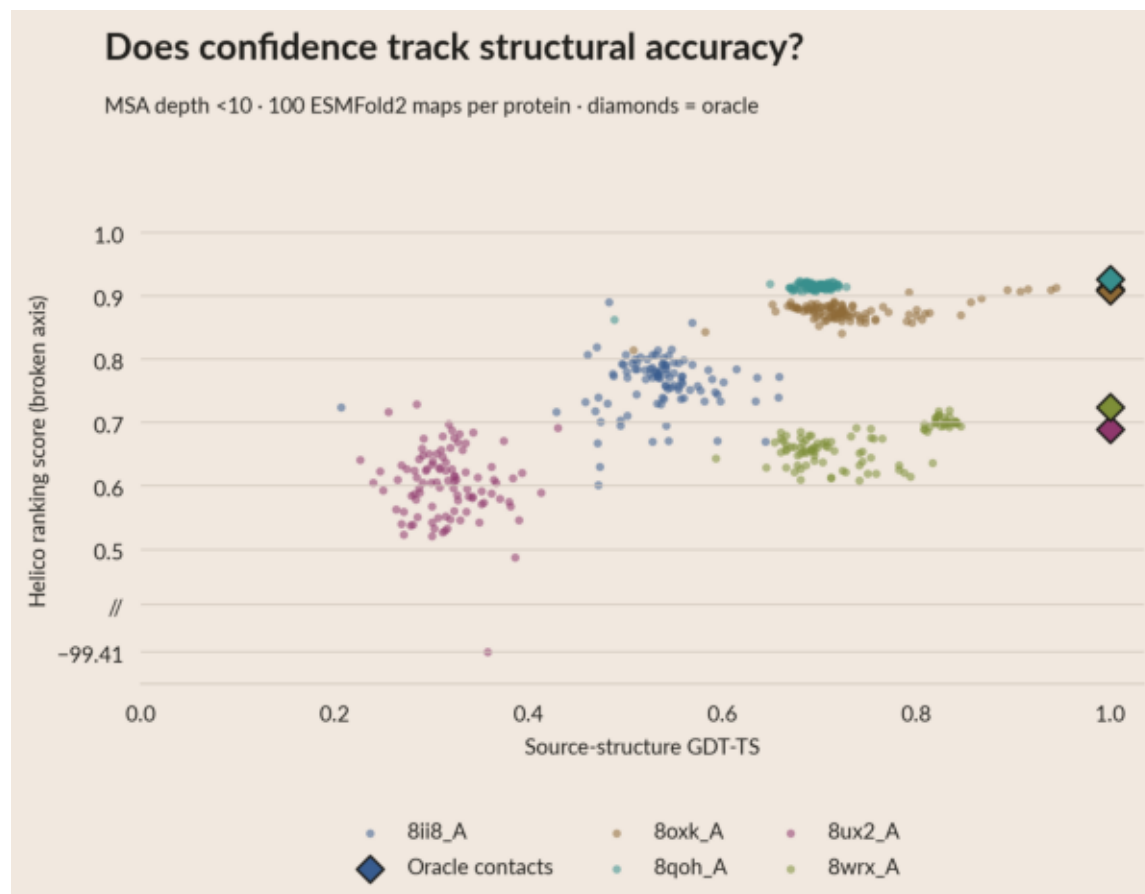
## 02b\_confidence.png

All five natural FoldBench proteins with MSA depth <10 (test split). Each dot is one of 100 single-sequence ESMFold2 predictions, converted to a full contact/non-contact map and folded by Helico; diamonds use the ground-truth oracle map. Every map has the same eligible-pair mask and three Helico diffusion samples, selected by ranking\_score. Contact counts may vary. Rank 1 is highest confidence among 101 maps; ties are shown as rank intervals. Vertical spread separates dots and has no numerical meaning. The menu switches to pLDDT of the same selected structures. These 500 decoys represent five proteins, not 500 independent biological examples. The ranking-score axis has an explicitly marked break to retain one clash-penalized score (−99.41) without compressing the other points; hover reports the unmodified score.



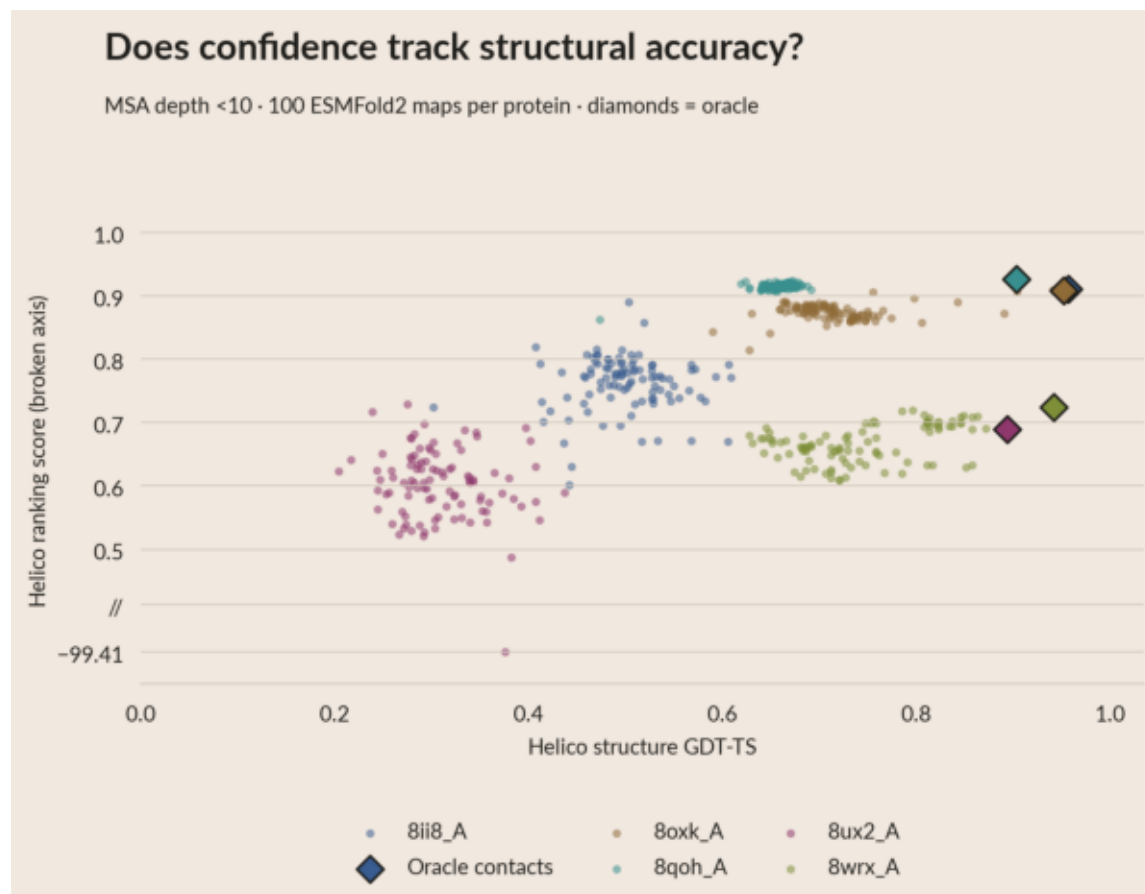
## 02c\_accuracy\_confidence.png

Original ESMFold2 structure accuracy versus Helico confidence after conditioning on its extracted contact map. Colors identify the five natural proteins with MSA depth <10; each has 100 seeded ESMFold2 predictions. Diamonds mark oracle contacts. In the source-structure view, oracle accuracy is 1 by definition because its source is the experimental reference. The menu can isolate each protein and also shows the measured accuracy of the resulting Helico structures, including the oracle-conditioned ones; GDT-TS and IDDT are separate options. Confidence always comes from the highest-ranking of three Helico samples per map. No seed is filtered by accuracy or confidence, and the 500 predictions represent only five proteins. The confidence axis has an explicitly marked break; the clash-penalized point at -99.41 is retained, and hover reports the unmodified score.



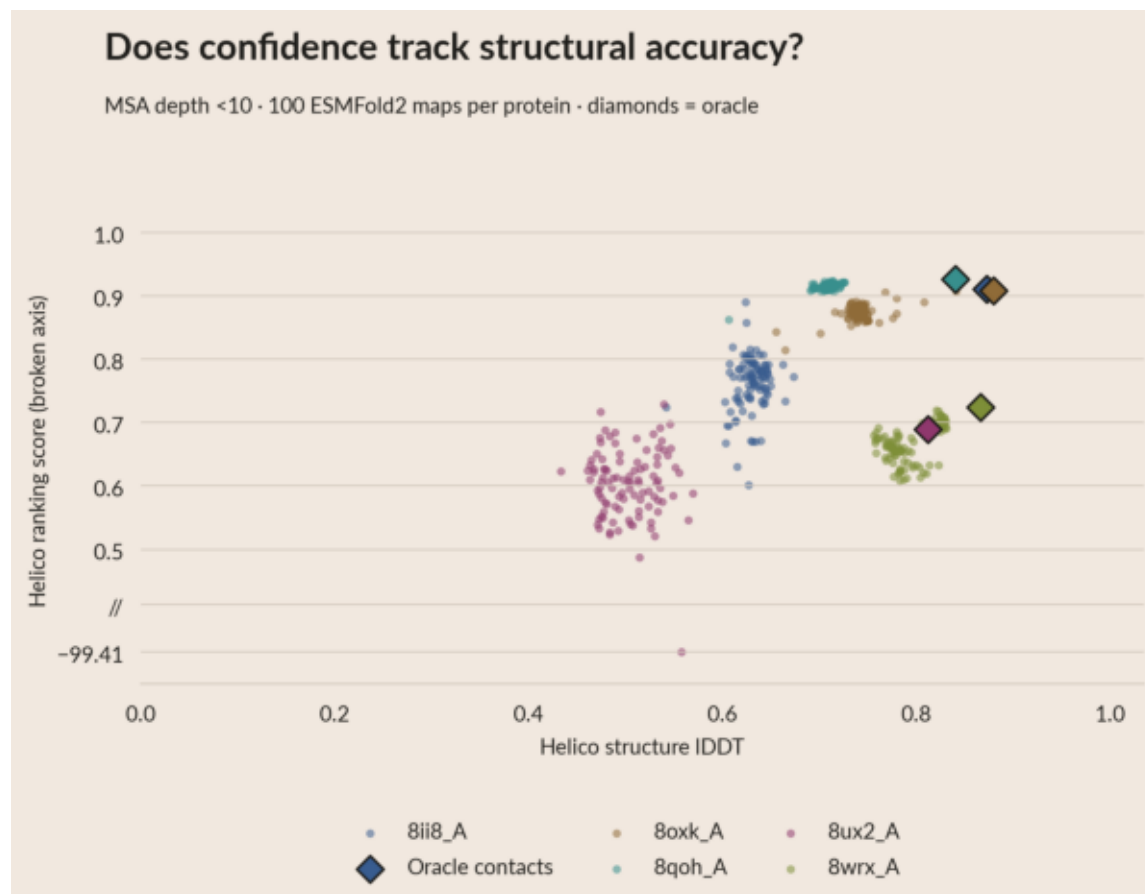
## 02c\_accuracy\_confidence\_gdt\_ts.png

Original ESMFold2 structure accuracy versus Helico confidence after conditioning on its extracted contact map. Colors identify the five natural proteins with MSA depth <10; each has 100 seeded ESMFold2 predictions. Diamonds mark oracle contacts. In the source-structure view, oracle accuracy is 1 by definition because its source is the experimental reference. The menu can isolate each protein and also shows the measured accuracy of the resulting Helico structures, including the oracle-conditioned ones; GDT-TS and IDDT are separate options. Confidence always comes from the highest-ranking of three Helico samples per map. No seed is filtered by accuracy or confidence, and the 500 predictions represent only five proteins. The confidence axis has an explicitly marked break; the clash-penalized point at -99.41 is retained, and hover reports the unmodified score.



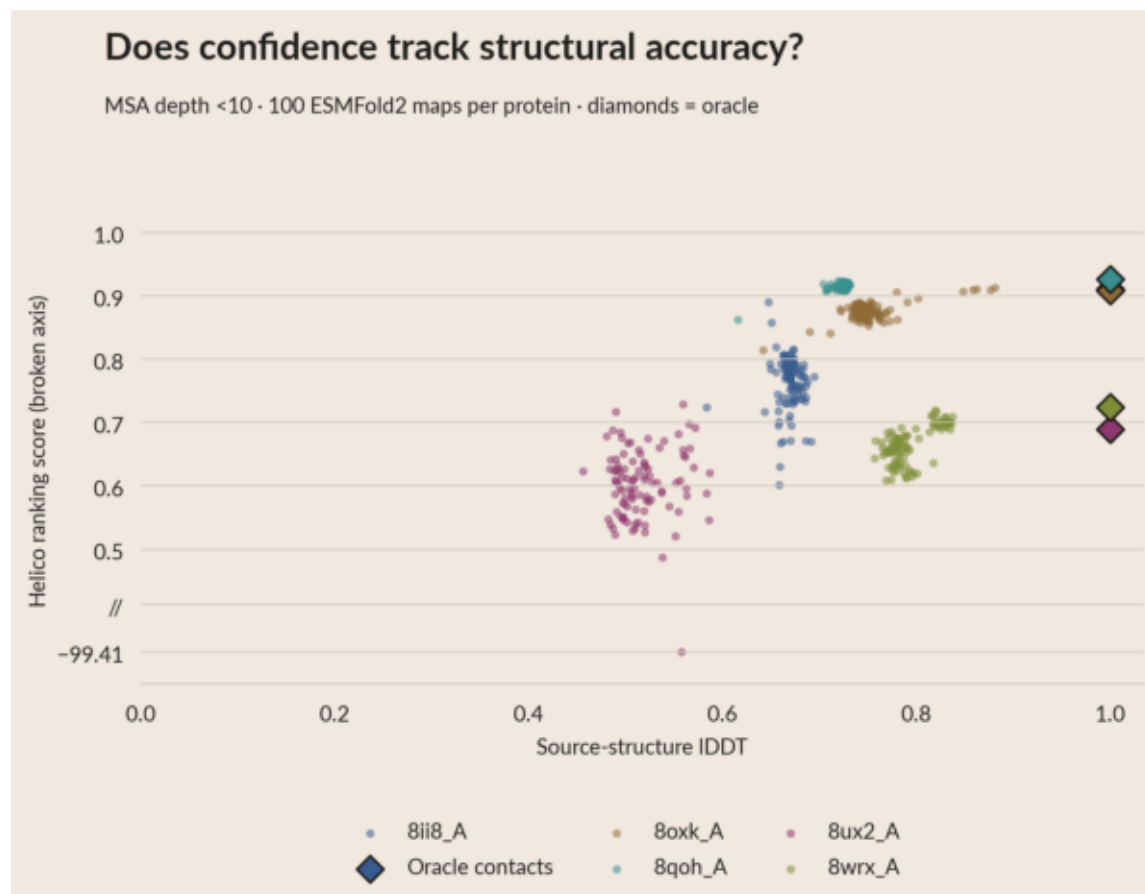
## 02c\_accuracy\_confidence\_lddt.png

Original ESMFold2 structure accuracy versus Helico confidence after conditioning on its extracted contact map. Colors identify the five natural proteins with MSA depth <10; each has 100 seeded ESMFold2 predictions. Diamonds mark oracle contacts. In the source-structure view, oracle accuracy is 1 by definition because its source is the experimental reference. The menu can isolate each protein and also shows the measured accuracy of the resulting Helico structures, including the oracle-conditioned ones; GDT-TS and IDDT are separate options. Confidence always comes from the highest-ranking of three Helico samples per map. No seed is filtered by accuracy or confidence, and the 500 predictions represent only five proteins. The confidence axis has an explicitly marked break; the clash-penalized point at -99.41 is retained, and hover reports the unmodified score.



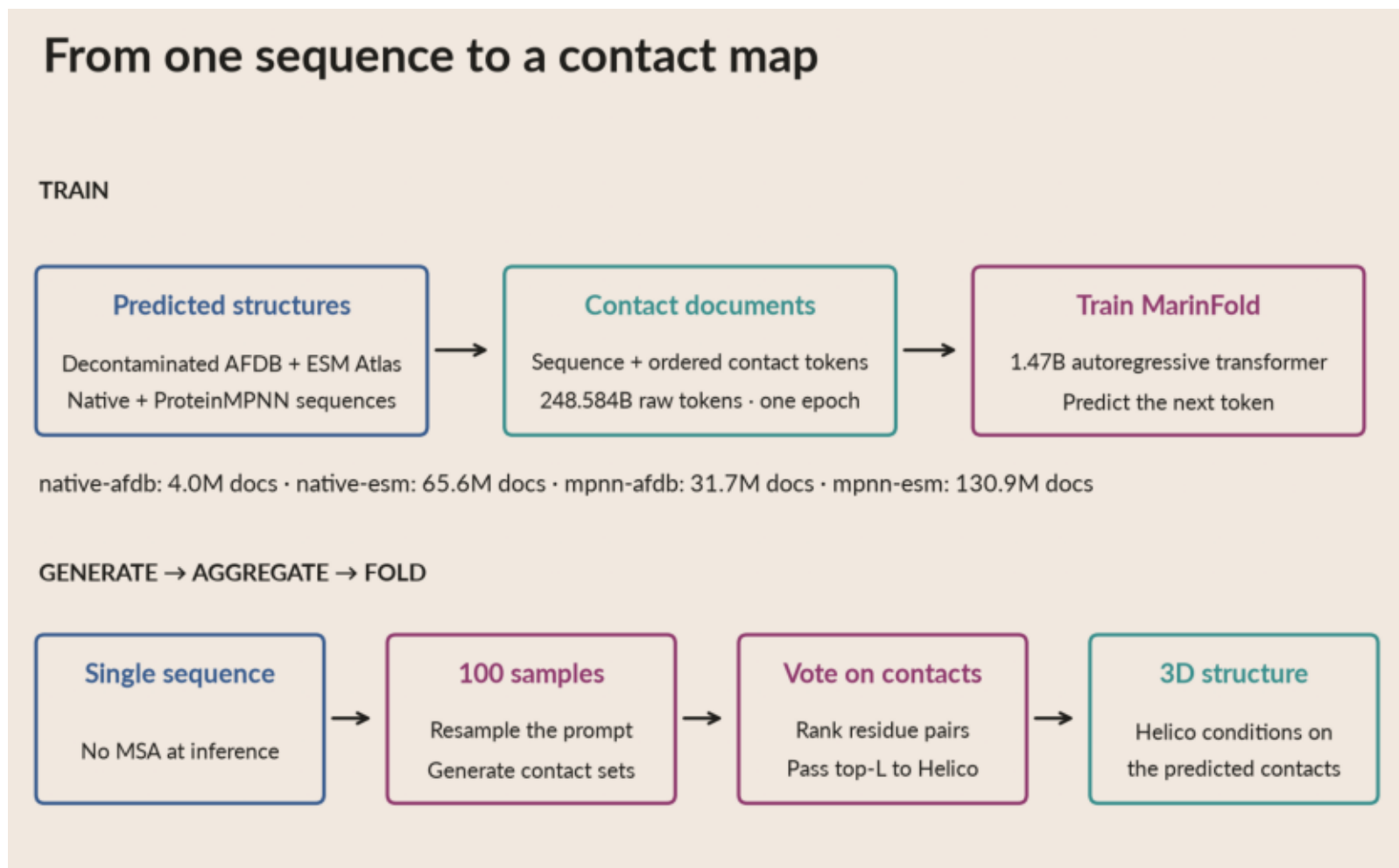
## 02c\_accuracy\_confidence\_source\_structure\_lddt.png

Original ESMFold2 structure accuracy versus Helico confidence after conditioning on its extracted contact map. Colors identify the five natural proteins with MSA depth <10; each has 100 seeded ESMFold2 predictions. Diamonds mark oracle contacts. In the source-structure view, oracle accuracy is 1 by definition because its source is the experimental reference. The menu can isolate each protein and also shows the measured accuracy of the resulting Helico structures, including the oracle-conditioned ones; GDT-TS and IDDT are separate options. Confidence always comes from the highest-ranking of three Helico samples per map. No seed is filtered by accuracy or confidence, and the 500 predictions represent only five proteins. The confidence axis has an explicitly marked break; the clash-penalized point at  $-99.41$  is retained, and hover reports the unmodified score.



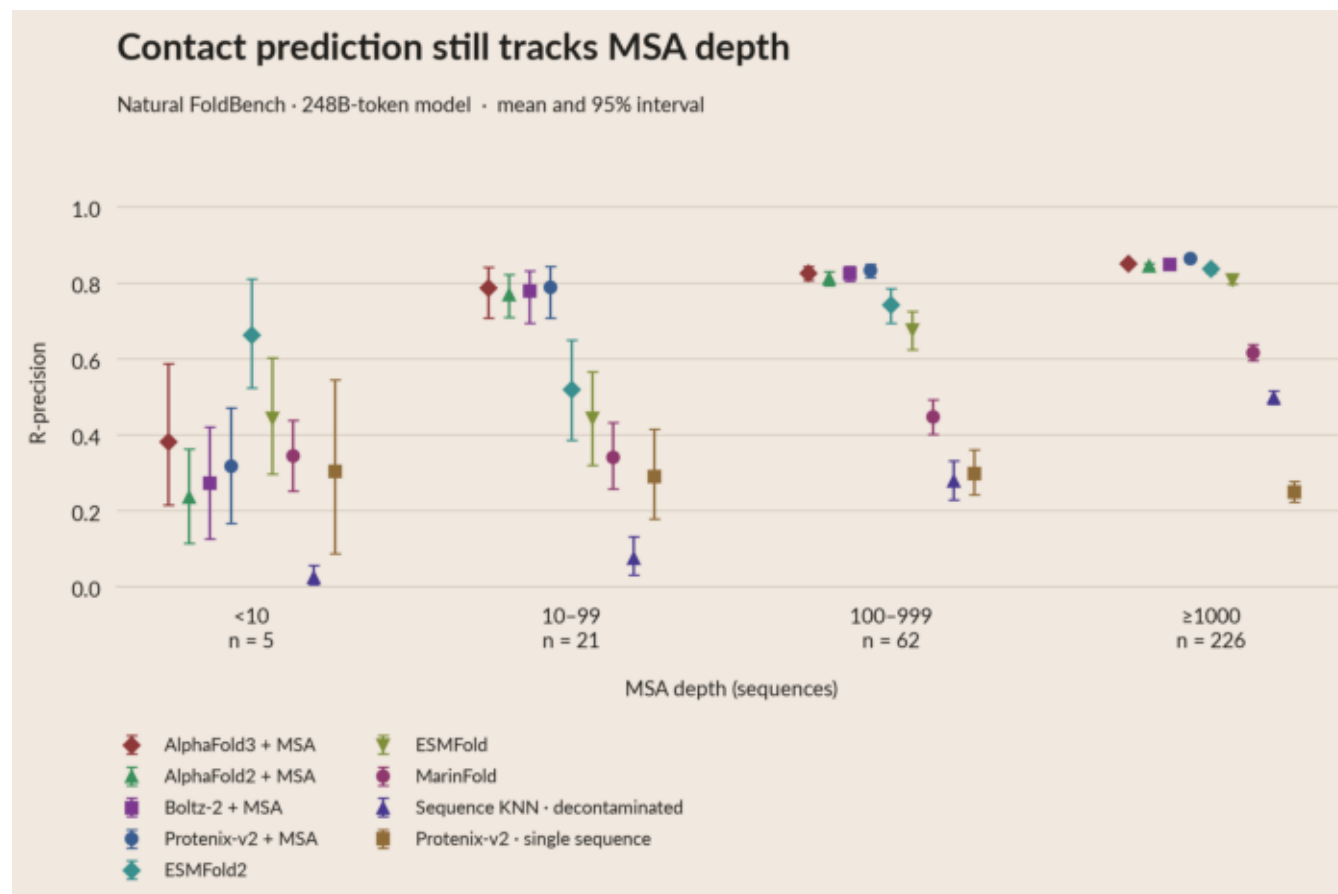
## 03\_method.png

Schematic; arrow widths do not encode amounts. Exp277 step 266,344: a scratch-trained 1.47B-parameter Qwen3, one epoch over 232,090,905 contact documents, totaling 248,583,762,834 raw tokens. Native and ProteinMPNN-redesigned sequences use AFDB / ESM Atlas source structures from the decontaminated corpus. Raw tokens differ from padded training slots.



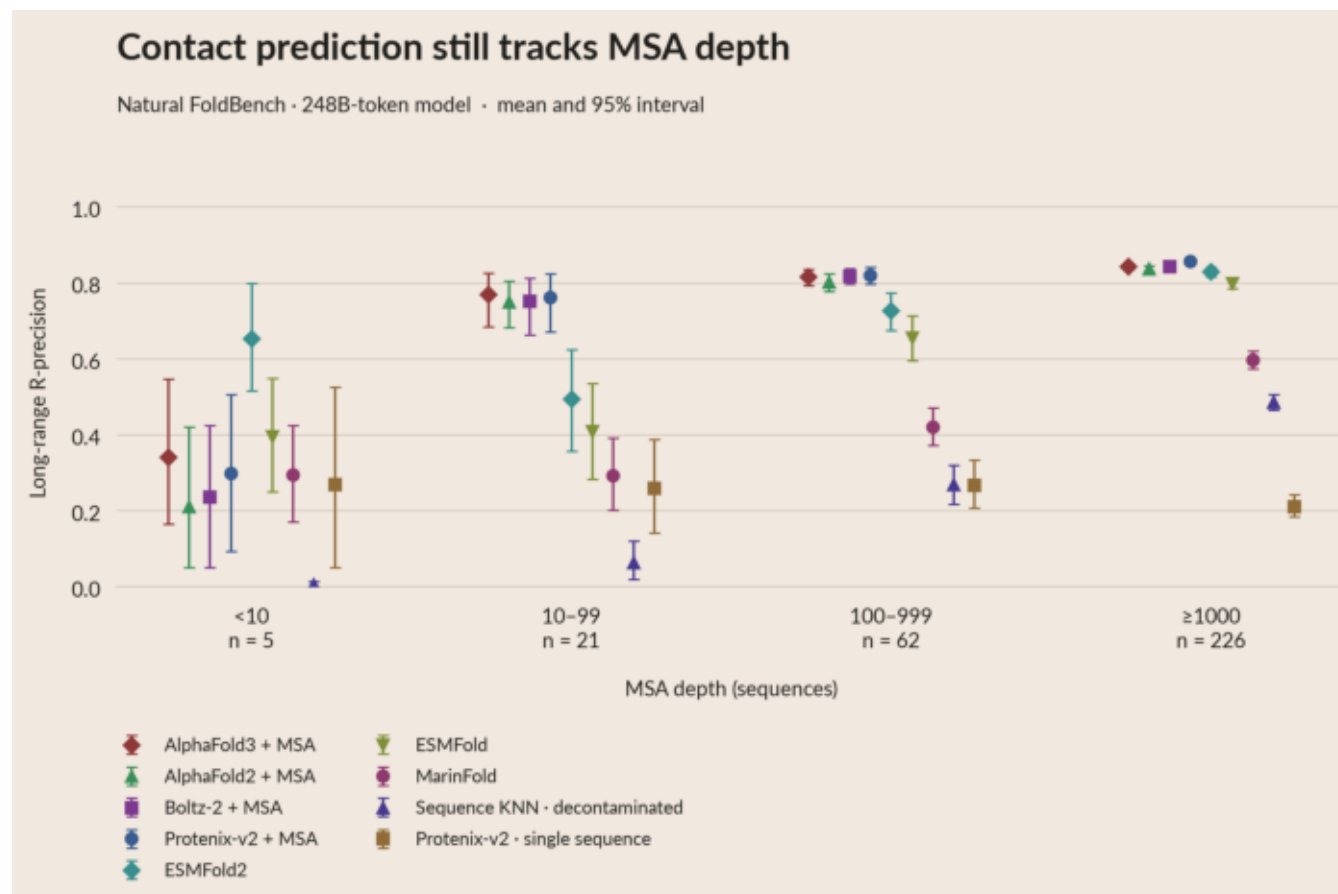
## 04\_contacts.png

Exp277 step 266,344, the 248.584B-raw-token model. All-range R-precision on 314 natural monomers (97 validation + 217 test); 100 resampled rollouts and vote ranking. R is the true-contact count in the resolved candidate universe. KNN indexes the native decontaminated corpus, not the additional redesign sequences. Only five natural proteins have depth <10. The menu also exposes long-range R-precision and separate split/design views. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



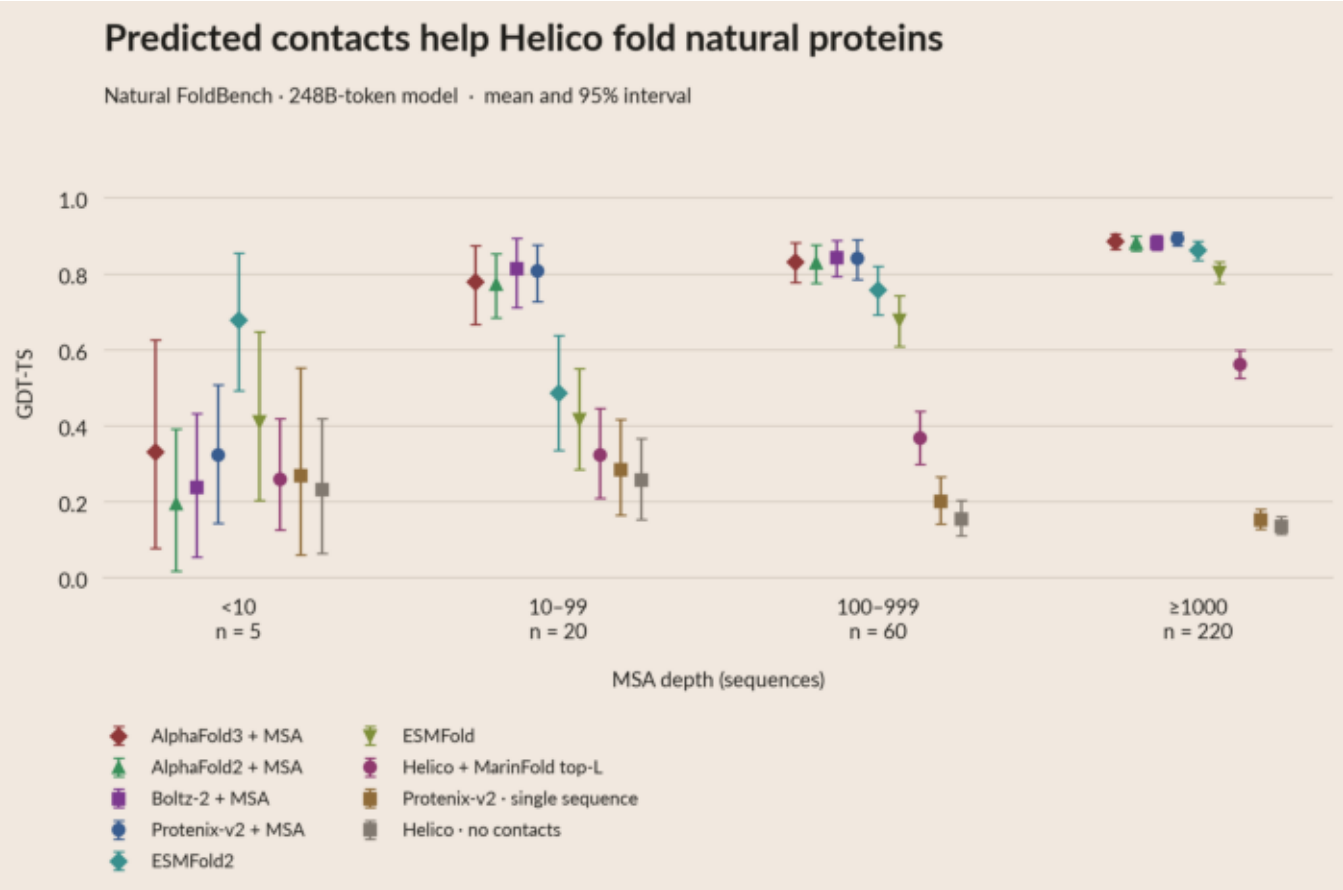
## 04\_contacts\_long.png

Exp277 step 266,344, the 248.584B-raw-token model. All-range R-precision on 314 natural monomers (97 validation + 217 test); 100 resampled rollouts and vote ranking. R is the true-contact count in the resolved candidate universe. KNN indexes the native decontaminated corpus, not the additional redesign sequences. Only five natural proteins have depth <10. The menu also exposes long-range R-precision and separate split/design views. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



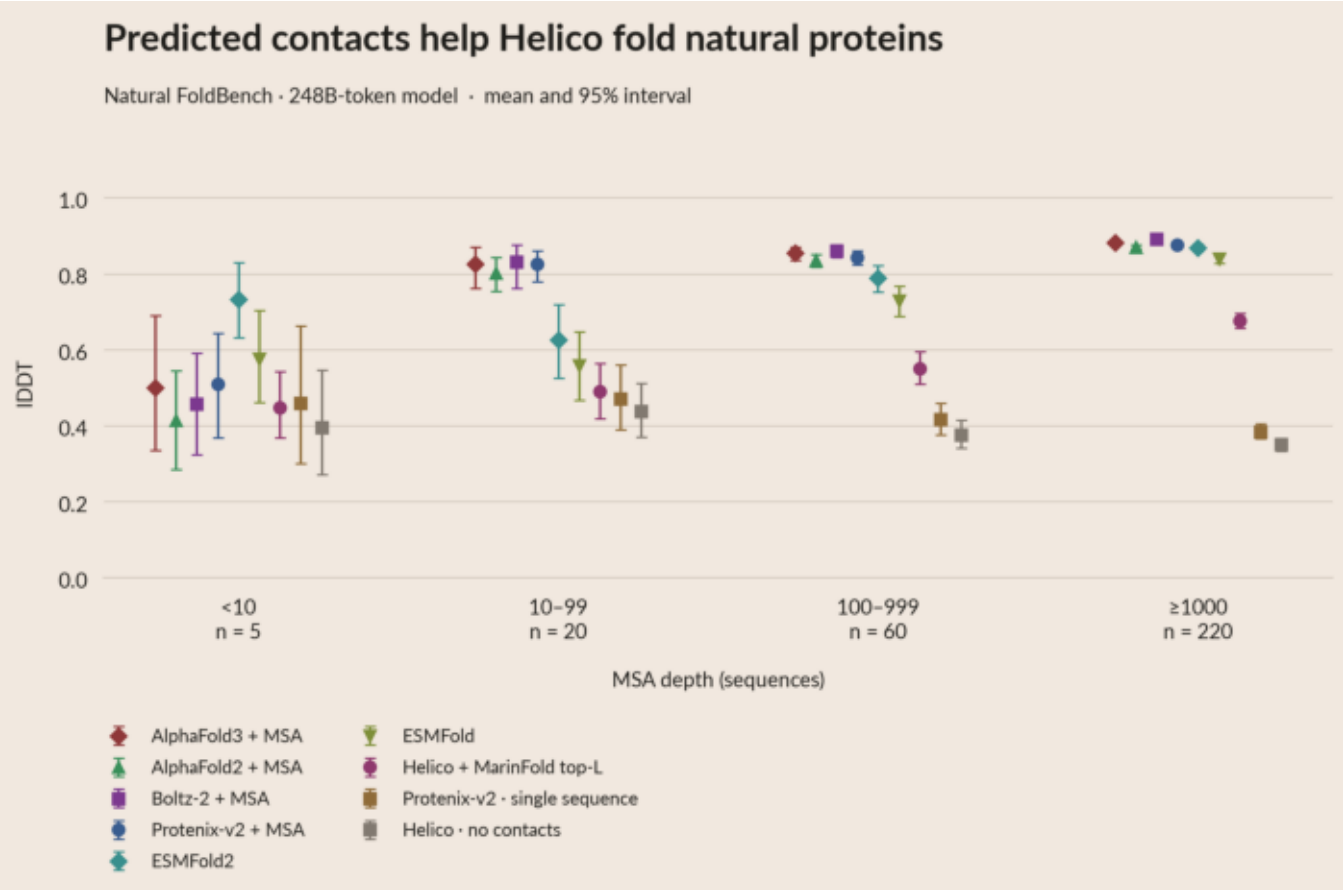
# 05\_folding.png

The 248B-token model through Helico: request the top-L predicted contacts, generate three diffusion samples, select by highest Helico confidence at that fixed cut; no MSA. Helico drops pairs closer than six residues after coordinate mapping; requested and effective counts are saved. Same 305 natural proteins in every arm (95 validation + 210 test). GDT-TS and IDDT are separate metrics. See coverage.csv for exclusions. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



# 05\_folding\_iddt.png

The 248B-token model through Helico: request the top-L predicted contacts, generate three diffusion samples, select by highest Helico confidence at that fixed cut; no MSA. Helico drops pairs closer than six residues after coordinate mapping; requested and effective counts are saved. Same 305 natural proteins in every arm (95 validation + 210 test). GDT-TS and IDDT are separate metrics. See coverage.csv for exclusions. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



## 06\_sampling.png

Exp277 step 266,344, 314 natural proteins (97 validation + 217 test), 100 iid rollouts per protein. Consensus and oracle use the same sample pool within each protein. Individual maps are ranked by emission order; short maps retain denominator R, and malformed or unfinished individual maps score zero. The diagnostic consensus uses all parsed maps, following exp321. Oracle selection uses ground truth. This measures contact accuracy, not distinct folds. Scatter colors identify MSA tiers in the interactive view.

